

Chapter 7 Notes: Bounded-Input Bounded-Output (BIBO) Stability

1. What Is BIBO Stability?

Bounded-Input Bounded-Output (BIBO) stability: for a *stable* system, feeding in a bounded input (e.g., an impulse or a step/setpoint) always produces a bounded output.

- A signal is **bounded** if there's some finite value its magnitude never exceeds.
- BIBO stability can be checked using the **open-loop transfer function** by looking at the roots of the **characteristic polynomial** — i.e., the **poles** of the system.

The core rule

- A pole is **stable** if its real part is **negative**.
- A system is **stable** only if **all** poles are stable.
- If **even one** pole is unstable, the **whole system is unstable**.

Big time-saver: you don't need to do the full inverse Laplace transform to check stability — just look at where the poles land. The worked examples below prove this by showing the time-domain answer matches what the pole locations predict.

2. Worked Example — Stable System

Given:

$$\frac{Y(s)}{R(s)} = \frac{s + 4}{s^2 + 3s + 2}$$

Time-domain check with an impulse, $r(t) = \delta(t)$:

$$Y(s) = \frac{s + 4}{(s + 1)(s + 2)} = \frac{k_1}{s + 1} + \frac{k_2}{s + 2} \Rightarrow y(t) = k_1 e^{-t} + k_2 e^{-2t}$$

$$y(\infty) = k_1 e^{-\infty} + k_2 e^{-\infty} = 0 \quad (\text{bounded})$$

Time-domain check with a step, $r(t) = 1$:

$$Y(s) = \frac{s + 4}{s(s + 1)(s + 2)} = \frac{k_1}{s} + \frac{k_2}{s + 1} + \frac{k_3}{s + 2} \Rightarrow y(t) = k_1 + k_2 e^{-t} + k_3 e^{-2t}$$

$$y(\infty) = k_1 \quad (\text{non-zero, but still bounded})$$

Pole-based shortcut (no time domain needed): set the characteristic polynomial to zero:

$$s^2 + 3s + 2 = (s + 1)(s + 2) = 0 \Rightarrow s = -1, -2$$

Both poles have negative real parts $\rightarrow$ **BIBO stable**. This matches the time-domain result exactly, without ever computing $y(t)$.

3. Worked Example — Unstable System

Given:

$$\frac{Y(s)}{R(s)} = \frac{s + 4}{s^2 + 3s - 2}$$

Time-domain check with an impulse:

$$Y(s) = \frac{s + 4}{(s + 1)(s - 2)} = \frac{k_1}{s + 1} + \frac{k_2}{s - 2} \Rightarrow y(t) = k_1 e^{-t} + k_2 e^{2t}$$

$$y(\infty) = k_1 e^{-\infty} + k_2 e^{\infty} \rightarrow \infty$$

The $k_1 e^{-t}$ term (from the stable pole at $s = -1$) decays fine on its own, but the $k_2 e^{2t}$ term (from the **unstable pole at $s = 2$**) blows up. **One unstable pole is enough to make the whole output unbounded.**

Pole-based shortcut:

$$s^2 + 3s - 2 = 0 \Rightarrow s = -1, 2$$

One pole ($s = 2$) has a **positive** real part $\rightarrow$ **BIBO unstable**.

4. Zeros vs. Poles — What Each One Controls

For an open-loop transfer function $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$:

	Comes from	Affects	Symbol on s-plane
Zeros	roots of $B(s)$ (numerator)	shape/dynamics of the response	"o"
Poles	roots of $A(s)$ (denominator)	shape/dynamics and stability	"x"

Zeros influence *how* the system responds (e.g., overshoot, shape of the curve) but **never determine stability** — only poles do that.

s-plane plots

For the stable example ($\frac{s+4}{(s+1)(s+2)}$): zero at $s = -4$ (o), poles at $s = -1, -2$ (x) — both poles left of the imaginary axis.

For the unstable example ($\frac{s+4}{(s+1)(s-2)}$): zero at $s = -4$ (o), poles at $s = -1$ (x, left of axis) and $s = 2$ (x, **right** of axis — this is what makes it unstable).

Visual rule of thumb: if *any* pole ("x") sits to the right of the imaginary axis (positive real part), the system is unstable.

5. Worked Example — Complex Poles (Stable)

Given:

$$\frac{Y(s)}{R(s)} = \frac{s+4}{s^2+6s+10}, \quad r(t) = \delta(t)$$

Factor into complex-conjugate poles: $s^2+6s+10 = (s+3+j)(s+3-j)$

$$Y(s) = \frac{k_1}{s+3+j} + \frac{k_2}{s+3-j}$$

Solve for residues using cover-up (same method as Chapter 6, just with complex substitution):

$$k_1 = \left. \frac{s+4}{s+3-j} \right|_{s=-3-j} = \frac{1-j}{-2j} = \frac{1+j}{2}$$

$$k_2 = \left. \frac{s+4}{s+3+j} \right|_{s=-3+j} = \frac{1+j}{2j} = \frac{1-j}{2}$$

Key observation: $k_2 = k_1^*$ (complex conjugate). This is always true — residues of complex-conjugate poles are always complex-conjugate pairs.

Converting to a real time-domain expression: using Euler's identities ($e^{-jt} = \cos t - j \sin t$, $e^{jt} = \cos t + j \sin t$) and combining terms, the imaginary parts cancel (because $k_1 + k_1^*$ is real and the $j \sin(t)$ coefficient becomes $k_1^* - k_1 = 0$... more precisely the sin terms cancel due to the conjugate symmetry), leaving:

$$y(t) = e^{-3t}(k_1 + k_1^*) \cos(t) = e^{-3t} \left(\frac{1+j}{2} + \frac{1-j}{2} \right) \cos(t) = e^{-3t} \cos(t)$$

Interpretation: a complex pole pair $-\sigma \pm j\omega$ produces a **decaying oscillation** $e^{-\sigma t} \cos(\omega t)$ in the time domain. The real part of the pole ($-\sigma$, here -3) controls the **decay envelope**; the imaginary part (ω , here 1) controls the **oscillation frequency**. Since the real part is negative, the oscillation decays to zero — stable.

6. Section 7.1 — Marginal Stability

Marginally stable poles sit *exactly* on the imaginary axis (real part = 0) — neither decaying nor growing on their own. They're trickier than the clear-cut stable/unstable cases.

Case A — Complex-conjugate poles on the imaginary axis

$$\frac{Y(s)}{R(s)} = \frac{k_1}{s + p_1} + \dots + \frac{k_n}{s + p_n} + \frac{b}{s^2 + b^2}$$

Here the binomial terms come from stable poles, and $\frac{b}{s^2 + b^2}$ comes from a complex-conjugate pair *on* the imaginary axis. With a step input ($r(t) = 1$):

$$y(t) = k_0 + k_1 e^{-p_1 t} + \dots + k_n e^{-p_n t} + \sin(bt)$$

As $t \rightarrow \infty$, all the stable-pole exponentials vanish, leaving:

$$y(\infty) = k_0 + \sin(bt)$$

This **oscillates forever but never grows unbounded** — so it's still **bounded** (BIBO stable), just perpetually oscillating rather than settling to a constant. This generalizes to multiple imaginary-axis pole pairs: each pair just adds another persistent $\sin(b_i t)$ term.

Case B — A single pole at the origin ($s = 0$)

This case is **input-dependent**:

- **Impulse input** ($R(s) = 1$): $Y(s) = \frac{k_1}{s} + \frac{k_2}{s+p_2} + \dots \Rightarrow y(t) = k_1 + k_2 e^{-p_2 t} + \dots$

$$y(\infty) = k_1 \quad (\text{bounded — BIBO stable for this input})$$

- **Step/setpoint input**: the pole at the origin combines with the $1/s$ from the step input itself, creating a **repeated pole at the origin** — i.e. an extra $\frac{k_m}{s^2}$ term:

$$y(t) = k_1 + k_2 e^{-p_2 t} + \dots + k_m t$$

$$y(\infty) = k_1 + k_m(\infty) \rightarrow \infty \quad (\text{unbounded!})$$

So a single pole at the origin is stable for an impulse but unstable for a step input — the classic example of why "marginal" cases need careful thought.

Summary rules for marginally-stable systems

1. **No root at the origin, but complex roots on the $j\omega$ axis**: stable — produces a bounded, oscillating output.
2. **A single root at the origin**: stability depends on the input — impulse $\rightarrow$ bounded; setpoint/step $\rightarrow$ unbounded.
3. **Repeated roots at the origin**: always unstable, regardless of input.

7. Section 7.2 — The Three Regions of the s-Plane

A pole is written as $s = \sigma + j\omega$, where σ = real part, $j\omega$ = imaginary part.

Region	Condition	Behavior
Stable region	$\sigma < 0$ (left half-plane)	Response decays to zero
Unstable region	$\sigma > 0$ (right half-plane)	Response grows unbounded
Marginally stable region	$\sigma = 0$ (on the $j\omega$ axis)	Response oscillates without growing or decaying (or is input-dependent at the origin)

Intuition: the imaginary part $j\omega$ produces oscillation (the $\cos / \sin$ term); the real part σ produces the exponential envelope ($e^{\sigma t}$) that either shrinks ($\sigma < 0$), grows ($\sigma > 0$), or stays constant ($\sigma = 0$) as $t \rightarrow \infty$.

8. Section 7.3 — Routh-Hurwitz Stability Criterion

Big idea: determine BIBO stability **without ever solving for the actual poles** — useful for high-order polynomials that are hard to factor, and essential for finding a *range* of an unknown design parameter that keeps a system stable.

Building the Routh Array

Given a characteristic polynomial of degree n :

$$A(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0$$

The Routh Array has $n + 1$ **rows**, labeled $s^n, s^{n-1}, \dots, s^0$.

First two rows come directly from the polynomial's coefficients, alternating:

Row	Col 1	Col 2	Col 3	...
s^n	a_n	a_{n-2}	a_{n-4}	...
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	...

Every row after that is calculated. **Each successive row loses one column** — the array shrinks to a single column by the final row (s^0).

The general formula for a calculated element x_{ij}

Build a 2×2 sub-matrix from the **first column** and the **column one to the right** of the target, using the two rows immediately above row i :

$$x_{ij} = \frac{- \begin{vmatrix} x_{(i-2)1} & x_{(i-2)(j+1)} \\ x_{(i-1)1} & x_{(i-1)(j+1)} \end{vmatrix}}{x_{(i-1)1}} = \frac{x_{(i-2)(j+1)} \cdot x_{(i-1)1} - x_{(i-2)1} \cdot x_{(i-1)(j+1)}}{x_{(i-1)1}}$$

In words: **cross-multiply the "outer" corners minus the "inner" corners of the 2×2 block above, then divide by the element directly above-left.** (Pad missing entries with zero.)

Stability rule from the array

Once complete, look **only at the first column**, top to bottom:

- **Count the sign changes.**
- The number of sign changes = the **number of unstable poles.**
- Zero sign changes $\rightarrow$ BIBO stable.

9. Routh Array Worked Examples

Example — Generic 3rd-order polynomial

$$A(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0:$$

$$x_{31} = \frac{a_2 a_1 - a_3 a_0}{a_2}, \quad x_{32} = 0 \text{ (padding with zero)}, \quad x_{41} = a_0$$

Final array:

s^3	a_3	a_1
s^2	a_2	a_0
s^1	x_{31}	0
s^0	a_0	

Example — $A(s) = s^2 - 3s - 4$

$$x_{31} = \frac{(1)(-4) - (-3)(0)}{-3} = \text{wait — using the formula: } - \begin{vmatrix} 1 & -4 \\ -3 & 0 \end{vmatrix} = \frac{-(12)}{-3} = -4$$

First column: $+1, -3, -4 \rightarrow$ **1 sign change** $\rightarrow$ **1 unstable pole.**

Example — $A(s) = s^2 + 5s + 4$

First column: $+1, +5, +4 \rightarrow$ **0 sign changes** $\rightarrow$ **stable, no unstable poles.**

Example — $A(s) = s^2 - 5s + 4$

First column: $+1, -5, +4 \rightarrow$ **2 sign changes** $\rightarrow$ **2 unstable poles** (for a 2nd-order system, this means **all** poles are unstable).

Example — Tunable Parameter (finding a stability range)

$A(s) = s^3 + 9s^2 + Ks + 12$, where K is an unknown design parameter.

s^3	1	K
s^2	9	12
s^1	$x_{31} = \frac{9K - 12}{9}$	$x_{32} = 0$
s^0	$x_{41} = 12$	

For stability, **every** first-column entry must stay positive (no sign changes). The only entry that depends on K is x_{31} :

$$\frac{9K - 12}{9} > 0 \Rightarrow 9K - 12 > 0 \Rightarrow K > \frac{12}{9} = 1\frac{1}{3}$$

Result: the system is stable for $1\frac{1}{3} < K < \infty$. This is a hugely practical use of Routh-Hurwitz — finding the safe range for a design/tuning parameter without ever solving the cubic.

10. Section 7.3.1 — Four Cases of Routh Array Analysis

1. **Nominal** — everything computes normally, no special handling needed (all examples above).
2. **Zero element in the first column** (but the rest of that row isn't all zero) — needs a limit-based workaround.
3. **A full row of zeros** — needs the "auxiliary equation" workaround.
4. **Indeterminate array** — the array simply can't be completed. In this case, **Routh-Hurwitz is sufficient to prove stability, but not necessary to prove instability** — i.e., failure to complete it doesn't tell you the system is unstable; you'd need another method.

Worked Example — Nominal (4th order, for practice)

$A(s) = s^4 + 16s^3 + 70s^2 + 196s - 816$. After computing all rows, the first column comes out:

$$+1, +16, +57.75, +422.08, -816$$

One sign change $\rightarrow$ 1 unstable pole (out of 4) $\rightarrow$ unstable overall.

Worked Example — Zero Element in the First Column

$$A(s) = s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10$$

When computing x_{31} , it comes out to exactly 0, but the *rest* of that row is non-zero (e.g. $x_{32} = 6 \neq 0$) — this is the "zero element" case, not the "row of zeros" case.

Fix: replace the zero with a small placeholder variable ε , finish computing the array symbolically in terms of ε , then take the limit as $\varepsilon \rightarrow 0^+$ for every element that depends on it.

Working through the algebra:

$$x_{31} = \lim_{\varepsilon \rightarrow 0^+} \varepsilon = +0, \quad x_{41} = \lim_{\varepsilon \rightarrow 0^+} \frac{4\varepsilon - 12}{\varepsilon} \rightarrow -\infty, \quad x_{51} = \lim_{\varepsilon \rightarrow 0^+} \frac{6x_{41} - 10\varepsilon}{x_{41}} = 6$$

Final first column: $+1, +2, +0, -\infty, +6, +10 \rightarrow$ **two sign changes $\rightarrow$ 2 unstable poles (of 5).**

Recipe for this case: (1) set the zero to ε ; (2) finish the array as normal, carrying ε through the algebra; (3) take $\lim_{\varepsilon \rightarrow 0^+}$ of each affected element; (4) count sign

| changes as usual.

11. Section 7.3.2 — Row of Zeros

When it happens: the characteristic polynomial has **poles symmetric about the origin** in the s-plane (e.g., a complex-conjugate pair exactly on the $j\omega$ axis, or real poles that are mirror images across the origin).

Fix — the Auxiliary Equation method:

- 1. When an entire row computes to all zeros, look at the row **directly above** it.
- 2. Build the **auxiliary polynomial** $P(s)$ using that row's coefficients as coefficients, with the row's own label (s^k) as the **highest power**, decreasing by 2 each term (since these come from *symmetric* poles).
- 3. Take $\frac{dP(s)}{ds}$.
- 4. Use the derivative's coefficients to **replace** the row of zeros.
- 5. Continue completing the array as normal.

Worked Example

$$A(s) = s^4 + 12s^3 + 36s^2 + 48s + 128$$

s^4	1	36	128
s^3	12	48	0
s^2	32	128	0
s^1	0	0	

Row s^1 is all zeros. Use the row above it (s^2 : coefficients 32, 128) to build the auxiliary equation, with highest power s^2 (matching the row label):

$$P(s) = 32s^2 + 128$$

Derivative:

$$\frac{dP(s)}{ds} = 64s + 0 \Rightarrow x_{41} = 64, x_{42} = 0$$

Completing the array gives first column $+1, +12, +32, +64, +128 \rightarrow$ **no sign changes** $\rightarrow$ **stable**.

Bonus insight: solving $P(s) = 0$ reveals the *actual* symmetric poles:

$$32s^2 + 128 = 0 \Rightarrow s^2 = -4 \Rightarrow s = \pm 2j$$

So two of the four poles sit exactly on the imaginary axis at $\pm 2j$ (marginally stable), and the other two are safely in the stable region. The Routh array alone told us "stable" but the auxiliary equation adds extra insight about *where* the poles are.

12. Section 7.3.3 — Indeterminate Array

If a characteristic polynomial produces an array that simply **cannot be completed**, this doesn't prove instability — the Routh test is **sufficient but not necessary** for proving stability. In this situation you'd need a different method (e.g., direct root-finding) to draw a conclusion.

13. Section 7.3.4 — Special Case: Multiple Valid Approaches

Some polynomials can be handled by *either* the "zero in first column" method *or* the "row of zeros" method, depending on how you set up the array. **Both are valid as long as you're consistent.**

Worked Example — $A(s) = s^3 - 9s$

Option 1 — Direct, with the auxiliary equation: Write $A(s) = s^3 + 0s^2 - 9s + 0$:

s^3	1	-9
s^2	0	0

Row s^2 is all zeros $\rightarrow$ auxiliary equation from the row **above** it isn't available in the usual sense here, so instead use $P(s) = A(s) = s^3 - 9s$ itself (matching the highest power to the current row's context), giving $\frac{dP}{ds} = 3s^2 - 9$. Replace the zero row: $s^2 \rightarrow 3, -9$.

Completing the array: first column $+1, +3, -6, -9 \rightarrow$ **one sign change** $\rightarrow$ **1 unstable pole**.

Option 2 — Factor out the common root first, then apply Routh to what's left:

$$A(s) = s(s^2 - 9)$$

The factored-out s itself represents a pole exactly at the origin (marginally stable by itself). Apply Routh only to $\hat{A}(s) = s^2 - 9 = s^2 + 0s - 9$:

s^2	1	-9
s^1	0	0

Row of zeros again $\rightarrow$ auxiliary equation $P(s) = s^2 - 9$, derivative $2s + 0$. Replace: $s^1 \rightarrow 2, 0$. Final first column: $+1, +2, -9 \rightarrow$ **unstable** (matches Option 1's conclusion).

Extra insight from Option 2: since $A(s) = s \cdot \hat{A}(s)$, we know the full system has:

- One pole exactly at the origin (marginally stable, from the factored-out s)
- Two poles symmetric about the origin, found by solving $P(s) = s^2 - 9 = 0 \Rightarrow s = \pm 3$ (one stable at -3 , one unstable at $+3$)

Lesson: when multiple valid approaches exist, they must agree — and sometimes reorganizing the polynomial (factoring out an obvious root) makes the analysis and its interpretation cleaner. **The key is to stay consistent and keep track of what you've already accounted for** (e.g., the factored-out pole).

Quick-Reference Summary

- **BIBO stability rule:** all poles must have negative real parts. One unstable pole $\rightarrow$ whole system unstable.
- **Zeros** shape the response; **poles** shape the response *and* determine stability.
- **s-plane regions:** $\sigma < 0$ stable, $\sigma > 0$ unstable, $\sigma = 0$ marginally stable.
- **Complex poles** $-\sigma \pm j\omega \rightarrow$ time response $e^{-\sigma t}(\cos / \sin(\omega t))$: envelope from σ , oscillation from ω .
- **Marginal stability special cases:**
 - Complex poles on $j\omega$ axis (no pole at origin) $\rightarrow$ bounded oscillation, stable.
 - Single pole at origin $\rightarrow$ depends on input (impulse: stable; step: unstable).
 - Repeated pole(s) at origin $\rightarrow$ always unstable.

- **Routh-Hurwitz:** build the array from $A(s)$'s coefficients; count sign changes in the **first column** = number of unstable poles.
- **Special array cases:**
 - *Zero in first column only* $\rightarrow$ replace with ε , finish the array, take $\lim_{\varepsilon \rightarrow 0^+}$.
 - *Entire row of zeros* $\rightarrow$ build the auxiliary equation $P(s)$ from the row above, differentiate, use those coefficients to replace the zero row. Roots of $P(s)$ reveal poles symmetric about the origin.
 - *Indeterminate array* $\rightarrow$ inconclusive; Routh-Hurwitz is sufficient but not necessary for proving stability.
- **Routh-Hurwitz is especially useful for finding the stable range of an unknown design parameter** (e.g., solving for the range of K that keeps a first-column term positive).